

Arithmetic and Edited Pictures

Arithmetic Verb Template

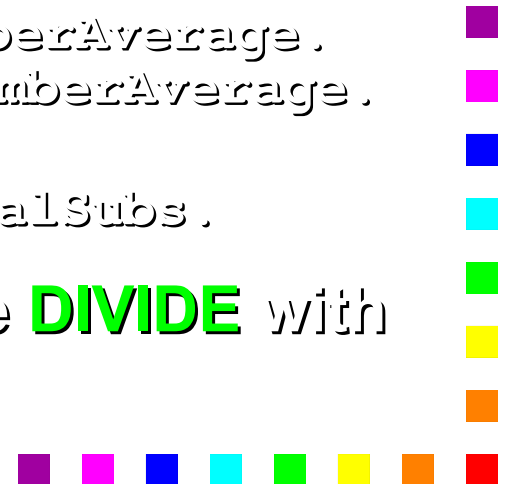
VERB { Identifier
Literal } { TO
FROM
BY
INTO } { Identifier ...
Identifier GIVING Identifier ... } [ROUNDED]

[ON SIZE ERROR StatementBlock END - VERB]

Most COBOL arithmetic verbs conform to the template above. For example;

ADD Takings TO CashTotal.
ADD Males TO Females GIVING TotalStudents.
SUBTRACT Tax FROM GrossPay.
SUBTRACT Tax FROM GrossPay GIVING NetPay.
DIVIDE Total BY Members GIVING MemberAverage.
DIVIDE Members INTO Total GIVING MemberAverage.
MULTIPLY 10 BY Magnitude.
MULTIPLY Members BY Subs GIVING TotalSubs.

The exceptions are the **COMPUTE** and the **DIVIDE** with REMAINDER.



The ROUNDED option

Receiving Field	Actual Result	Truncated Result	Rounded Result
PIC 9(3)V9.	123.25	123.2	123.3
PIC 9(3).	123.25	123	123

The ROUNDED option takes effect when, after decimal point alignment, the result calculated must be truncated on the right hand side.

The option adds 1 to the receiving item when the leftmost truncated digit has an absolute value of 5 or greater.

The ON SIZE ERROR option

Receiving Field	Actual Result	SIZE ERROR
PIC 9(3)V9.	245.9 6	Yes
PIC 9(3)V9.	1 245.9	Yes
PIC 9(3).	124	No
PIC 9(3).	1 246	Yes
PIC 9(3)V9 Not Rounded	124.4 5	Yes
PIC 9(3)V9 Rounded	124.4 5	No
PIC 9(3)V9 Rounded	3 124.4 5	Yes

A size error condition exists when, after decimal point alignment, the result is truncated on either the **left** or the **right** hand side.

If an arithmetic statement has a rounded phrase then a size error only occurs if there is truncation on the **left** hand side (most significant digits).

ADD Examples

ADD Cash TO Total.

Before
After

3	1000
3	1003

ADD Cash, 20 TO Total, Wage.

Before
After

3	1000	100
3	1023	123

ADD Cash, Total GIVING Result.

Before
After

3	1000	0015
3	1000	1003

ADD Males TO Females GIVING TotalStudents.

Before
After

1500	0625	1234
1500	0625	2125

SUBTRACT Examples

SUBTRACT Tax FROM GrossPay, Total.

Before

After

120
120

4000
3880

9120
9000

SUBTRACT Tax, 80 FROM Total.

Before

After

100
100

480
300

SUBTRACT Tax FROM GrossPay GIVING NetPay.

Before

After

750
750

1000
1000

0012
0250

MULTIPLY and DIVIDE Examples

MULTIPLY Subs BY Members GIVING TotalSubs
ON SIZE ERROR DISPLAY "TotalSubs too small"
END-MULTIPLY.

Before
After

Subs	Members	TotalSubs
15.50	100	0123.45
15.50	100	1550.00

MULTIPLY 10 BY Magnitude, Size.

Before
After

355	125
3550	1250

DIVIDE Total BY Members GIVING Average ROUNDED.

Before
After

9234.55	100	1234.56
9234.55	100	92.35

The Divide Exception

DIVIDE { Identifier
Literal } INTO { Identifier
Literal } GIVING { Identifier [ROUNDED] } REMAINDER Identifier
[{ ON SIZE ERROR
NOT ON SIZE ERROR } StatementBlock END - DIVIDE]

DIVIDE { Identifier
Literal } BY { Identifier
Literal } GIVING { Identifier [ROUNDED] } REMAINDER Identifier
[{ ON SIZE ERROR
NOT ON SIZE ERROR } StatementBlock END - DIVIDE]

DIVIDE 201 BY 10 GIVING Quotient REMAINDER Remain.

Before

After

209

020

424

001

The COMPUTE

COMPUTE { Identifier [ROUNDED] } ... = ArithmeticExpression

[{ ON SIZE ERROR
NOT ON SIZE ERROR } StatementBlock END - COMPUTE]

Precedence Rules.

1. ** = POWER N^N
2. * = MULTIPLY $\times$
/ = DIVIDE $\div$
3. + = ADD $+$
- = SUBTRACT $-$

Compute IrishPrice = SterlingPrice / Rate * 100.

Before
After

1000.50

179.59

156.25

156.25

87

87

Edited Pictures.

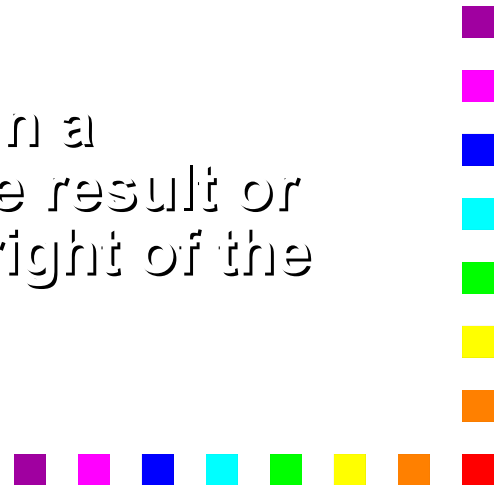
Edited Pictures are PICTURE clauses which **format** data intended for output to screen or printer.

To enable the data items to be formatted in a particular style COBOL provides additional picture symbols supplementing the basic **9**, **X**, **A**, **V** and **S** symbols.

The additional symbols are referred to as “Edit Symbols” and PICTURE clauses which include edit symbols are called “Edited Pictures”.

The term edit is used because the edit symbols have the effect of changing, or editing, the data inserted into the edited item.

Edited items can **not** be used as operands in a computation but they may be used as the result or destination of a computation (i.e. to the right of the word GIVING).



Editing Types

COBOL provides two basic types of editing

① Insertion Editing - which modifies a value by including additional items.

② Suppression and Replacement Editing - which suppresses and replaces leading zeros.

Each type has sub-categories

- Insertion editing
 - ✴ Simple Insertion
 - ✴ Special Insertion
 - ✴ Fixed Insertion
 - ✴ Floating Insertion
- Suppression and Replacement
 - ✴ Zero suppression and replacement with spaces
 - ✴ Zero suppression and replacement with asterisks



Editing Symbols

Edit Symbol	Editing Type
, B 0 /	Simple Insertion
.	Special Insertion
+ - CR DB \$	Fixed Insertion
+ - S	Floating Insertion
Z *	Suppression and Replacement



Simple Insertion.

Sending		Receiving	
Picture	Data	Picture	Result
PIC 999999	123456	PIC 999,999	123,456
PIC 9(6)	000078	PIC 9(3),9(3)	000,078
PIC 9(6)	000078	PIC ZZZ,ZZZ	78
PIC 9(6)	000178	PIC ***,**	***178
PIC 9(6)	002178	PIC ***,**	**2,178
PIC 9(6)	120183	PIC 99B99B99	12 01 83
PIC 9(6)	120183	PIC 99/99/99	12/01/83
PIC 9(6)	001245	PIC 990099	120045

Special Insertion.

Sending		Receiving	
Picture	Data	Picture	Result
PIC 999V99	12345	PIC 999.99	123.45
PIC 999V99	02345 	PIC 999.9	023.4
PIC 999V99	51234 	PIC 99.99	12.34
PIC 999	456. 	PIC 999.99	456.00
			

Fixed Insertion - Plus and Minus.

Sending		Receiving	
Picture	Data	Picture	Result
PIC S999	-123	PIC -999	-123
PIC S999	-123	PIC 999-	123-
PIC S999	+123	PIC -999	123
PIC S9(5)	+12345	PIC +9(5)	+12345
PIC S9(3)	-123	PIC +9(3)	-123
PIC S9(3)	-123	PIC 999+	123-

Fixed Insertion - Credit, Debit, \$

Sending		Receiving	
Picture	Data	Picture	Result
PIC S9(4)	+1234	PIC 9(4) CR	1234
PIC S9(4)	-1234	PIC 9(4) CR	1234CR
PIC S9(4)	+1234	PIC 9(4) DB	1223
PIC S9(4)	-1234	PIC 9(4) DB	1234DB
PIC 9(4)	1234	PIC \$ 99999	\$01234
PIC 9(4)	0000	PIC \$ ZZZZZ	\$

Floating Insertion.

Sending		Receiving	
Picture	Data	Picture	Result
PIC 9(4)	0000	PIC \$\$,\$\$9.99	\$0.00
PIC 9(4)	0080	PIC \$\$,\$\$9.00	\$80.00
PIC 9(4)	0128	PIC \$\$,\$\$9.99	\$128.00
PIC 9(5)	57397	PIC \$\$,\$\$9	\$7,397
PIC S9(4)	- 0005	PIC ++++9	-5
PIC S9(4)	+0080	PIC ++++9	+80
PIC S9(4)	- 0080	PIC - - - - 9	-80
PIC S9(5)	+71234	PIC - - - - 9	ž1234

Suppression and Replacement

Sending		Receiving	
Picture	Data	Picture	Result
PIC 9(5)	12345	PIC ZZ,999	12,345
PIC 9(5)	01234	PIC ZZ,999	1,234
PIC 9(5)	00123	PIC ZZ,999	123
PIC 9(5)	00012	PIC ZZ,999	012
PIC 9(5)	05678	PIC **, **9	*5,678
PIC 9(5)	00567	PIC **, **9	***567
PIC 9(5)	00000	PIC **, ***	*****